Exercises

Problem 3.1
Prove Eqs. (3.49) and (3.50) by using the Campbell identity.

Problem 3.2
Prove the conservation of the commutation relation for rotated quadratures, Eq. (3.55).

Problem 3.3: Mean and variance under displacements
Consider a quantum harmonic ocillator in state $|\psi\rangle$. Show that a displacement $|\psi\rangle \to \hat{D}(\alpha)|\psi\rangle$ transforms the expectation values of $\hat{x}$ and $\hat{p}$ as

$$\langle\hat{x}\rangle_\psi = \langle\psi|\hat{x}|\psi\rangle \to \langle\hat{x}\rangle_\psi + \sqrt{2}\,\mathrm{Re}(\alpha),$$
$$\langle\hat{p}\rangle_\psi = \langle\psi|\hat{p}|\psi\rangle \to \langle\hat{p}\rangle_\psi + \sqrt{2}\,\mathrm{Im}(\alpha).$$

Show that the variances $\langle\Delta^2\hat{x}\rangle$ and $\langle\Delta^2\hat{p}\rangle$ remain unchanged.

Problem 3.4: Reality of the Wigner function
Show that the Wigner function as defined by Eq. (3.80) is real.

Problem 3.5: Wigner function marginals
Show that for pure states, the marginals of the Wigner function equal the probability densities given by the $\hat{x}$- and $\hat{p}$-space wave function as in Eq. (3.83).

Problem 3.6: Single-mode squeezing transformation
Show that the annihilation operator transforms under single-mode squeezing as given by Eq. (3.97).

Problem 3.7: Two-mode squeezing transformation
Show that the annihilation operators for a pair of modes transform under two-mode squeezing as given by Eqs. (3.109) and (3.110).

Problem 3.8: Beam splitter between coherent states
Prove Eq. (3.141), showing that products of coherent states remain products when mixed on beam splitters.

Problem 3.9: Implementing displacements
Show that the construction described in Sec. 3.2.2 and Fig. 3.13 realises a displacement for arbitrary input states. *Hint: you can use the P-representation.*

Problem 3.10: Coherent state subject to loss
A coherent state $|\alpha\rangle$ is sent through a lossy optical channel with transmissivity η. What is the output state? What is the mean number of photons detected from a coherent state $|\alpha\rangle$ by a photon counter with efficiency η (and no dark counts)?